Institute of Technology of Cambodia
ECAM LaSalle

Name:	**Sovan Chhorponn**
ID:	**e20220873**

Network and Security
Final Exam (2 hours)

I. Questions

1. (10 pts) How many layers are there in network communication?
2. (10 pts) What is the difference between stateless and stateful inspection firewall?
3. (10 pts) How many types of threat are there?
4. (10 pts) What is the difference between Two and Three Tier network design?
5. (10 pts) What is the functionality of AI-driven Threat Detection?

II. Exercise

1. (20 points) Assume that we need to design network infrastructure for one startup company and there are 5 departments. Each departments needs available hosts 30, 56, 10, 45, 64. Make IP calculation for this start up company.
2. (30 Points) If we assume that the result of expansion box is:
   - Results of expansion box:
     001111100000111111100000001111111101110001100011
   - The schedule key is:
     101010100101010111011010001011100000110101010101
   - The provided fix table:

[5,	6,	1,	13,	14,	0,	6,	12,	1,	15,	7,	9,	14,	2,	7,	3]
[2,	3,	1,	1,	1,	3,	1,	11,	7,	9,	3,	5,	15,	14,	8,	13]
[9,	0,	13,	11,	7,	15,	15,	3,	13,	9,	13,	12,	7,	14,	15,	8]
[4,	12,	4,	12,	13,	6,	13,	2,	14,	4,	6,	8,	14,	1,	9,	8]

⇨ What will be the result of f functions, if the same fix table is used for all S-Box Operation?

I. Questions

1. We have 2 model OSI model (7 layers) or TCP/IP model (4 layers). In this case we focus on the OSI model (counting from number 7 to number 1).

   - Number 7: Application layer: It is responsible for providing the interface to the application user. It works with different protocol such as: HTTP, HTTPS, FTP, SSH, DNS…

   - Number 6: Presentation layer: It is known as a translation layer. It defines the data format and encryption. As it focusses on how to format data it is also called the syntax layer. (Compress & Decompress, Encryption and Decryption). It works with different protocol such as: LPP, SSL, SSH, NDR…

   - Number 5: Session layer: It establishes, monitors and terminates the communication session between applications. It makes a checkpoints to make data synchronization and dialog control (half-duplex or full-duplex). It works with different function such as NetBIOS, NFS (Network File System)…

   - Number 4: Transport layer: It is responsible for message delivery between the networked hosts. Messages are fragmented and reassembled by this layer. It also controls the reliability of any given link. It works with different protocol such as TCP:(Transmission Control Protocol (Connection-Oriented Service), UDP :User Datagram Protocol (Connectionless service)

   - Number 3: Network layer: It provides the routing process and logical addressing. It works with different protocol such as Routing protocols (RIP, OSPF…)

   - Number 2: Data link layer: It is framing, physical addressing, Error control, Flow control and access control. It works with different protocol such as Switching Protocol(ARP, ATM…)

   - Number 1: It is the last layer of OSI Model that deals with transmitting raw bits of data over a physical medium.

2. Difference between Stateless inspection firewall and Stateful inspection firewall:

   - Stateless inspection firewall: treats every packet as an entirely new, individual transaction. It makes a decision to allow or deny the packet based only on a static list of rules, called an Access Control List (ACL), and the information in the packet's header.
   - Stateful inspection firewall: maintains a state table that tracks the characteristics of every active connection, such as source/destination IP and port, and the sequence of packets (especially for TCP).

3. There are 4 types of threats:

   They are:

   - Structured Theat
   - Unstructured Threat
   - Insider
   - Outsider

   These types of threat have 3 sources: Human, environment and technology.

   - Human: Intentional and Unintentional
   - Environment and Technology: Unintentional

4. Difference between tier 2 and tier 3 infrastructure network design:
    - Tier 2 network infrastructure that we combine core network layer and distribution network layer to be a single layer. Having 2 main layers.
    - Tier 3 network infrastructure: has 3 layers like a normal infrastructure but the source is all duplicated or double.
5. AI driven threat detection is intrusion detection system that deployed the ai driven technology allowing the AI to learn, analyse output, upgrade the threat protection. There are 4 such as Predictive Analytic, Real time threat analysis, Anomaly detection, and automating routine tasks.

II. Exercise

1. Let's make IP calculation:

Department in descending order:

Department 5: 64 host < $2^7 - 2$

Department 2: 56 host < $2^6 - 2$

Department 4: 45 host < $2^6 - 2$

Department 1: 30 host < $2^5 - 2$

Department 3: 10 host < $2^4 - 2$

- Since the total is bigger than 2^8 and smaller than 2^{16}

Choose private IP address of 172.16.0.0/16

<u>Department 5</u>: 64 host < $2^7 - 2$

IP: $2^{32-subnetmask}$= 2^7 = 128

Subnetmask = 25

Subnetwork = $2^{25-16} = 2^9$

We get Department 5

Subnet 0 =$IP_{net} = 172.16.0.0\ /25$

$IP_{usable} = 172.16.0.1\ /25 \ \rightarrow \ 172.16.0.126\ /25$

$IP_{broadcast} = 172.16.0.127\ /25$

Subnet 1 =$IP_{net} = 172.16.0.128\ /25$

$IP_{usable} = 172.16.0.1\ /25 \ \rightarrow \ 172.16.0.254\ /25$

$IP_{broadcast} = 172.16.0.255\ /25$

Subnet 2 =$IP_{net} = 172.16.1.0\ /25$

$IP_{usable} = 172.16.1.1\ /25 \ \rightarrow \ 172.16.1.126\ /25$

$IP_{broadcast} = 172.16.1.127\ /25$

… … … … … … … … … … … … …

Subnet 511 =$IP_{net} = 172.16.255.128\ /25$

$IP_{usable} = 172.16.255.129\ /25 \ \rightarrow \ 172.16.255.254\ /25$

$IP_{broadcast} = 172.16.255.255\ /25$

<u>Department 2</u>: 56 host < $2^6 - 2$ Using Subnet 1 =$IP_{net} = 172.16.0.128\ /25$

IP: $2^{32-subnetmask}$= 2^6 = 64

Subnetmask = 26

Subnetwork = $2^{26-25} = 2^1 = 2$

We get Department 2

Subnet_0 =$IP_{net} = 172.16.0.128\ /26$

$IP_{usable} = 172.16.0.129\ /26 \ \rightarrow \ 172.16.0.190\ /26$

$IP_{broadcast} = 172.16.0.191\ /26$

Subnet_1 =$IP_{net} = 172.16.0.192\ /26$

$IP_{usable} = 172.16.0.193\ /26 \ \rightarrow \ 172.16.0.254\ /26$

$IP_{broadcast} = 172.16.0.255\ /26$

Department 4: 45 host < $2^6 - 2$ Using Subnet_1 =$IP_{net} = 172.16.0.192\ /26$

IP: $2^{32-subnetmask}$= 2^6 = 64

Subnetmask = 26

Subnetwork = $2^{26-26} = 2^0 = 1$

We get Department 4

Subnet_1 =$IP_{net} = 172.16.0.192\ /26$

$IP_{usable} = 172.16.0.193\ /26 \rightarrow 172.16.0.254\ /26$

$IP_{broadcast} = 172.16.0.255\ /26$

Department 1: 30 host < $2^5 - 2$ Using Subnet 2 =$IP_{net} = 172.16.1.0\ /25$

IP: $2^{32-subnetmask}$= 2^5 = 32

Subnetmask = 27

Subnetwork = $2^{27-25} = 2^2 = 4$

We get Department 1

Subnet-0 =$IP_{net} = 172.16.1.0\ /27$

$IP_{usable} = 172.16.1.29\ /27 \rightarrow 172.16.1.30\ /27$

$IP_{broadcast} = 172.16.1.31\ /27$

Subnet-1 =$IP_{net} = 172.16.1.32\ /27$

$IP_{usable} = 172.16.1.33\ /27 \rightarrow 172.16.1.62\ /27$

$IP_{broadcast} = 172.16.1.63\ /27$

Subnet-2 =$IP_{net} = 172.16.1.64\ /27$

$IP_{usable} = 172.16.1.65\ /27 \rightarrow 172.16.1.94\ /27$

$IP_{broadcast} = 172.16.1.95\ /27$

Subnet-3 =$IP_{net} = 172.16.1.96\ /27$

$IP_{usable} = 172.16.1.97\ /27 \rightarrow 172.16.1.126\ /27$

$IP_{broadcast} = 172.16.1.127\ /27$

Department 3: 10 host < $2^4 - 2$ Using Subnet-1 =$IP_{net} = 172.16.1.32\ /27$

IP: $2^{32-subnetmask}$= 2^4 = 16

Subnetmask = 28

Subnetwork = $2^{28-27} = 2^1 = 2$

We get Department 3

Subnet/0 = $IP_{net} = 172.16.1.32\ /28$

$IP_{usable} = 172.16.1.33\ /28 \rightarrow 172.16.1.46\ /28$

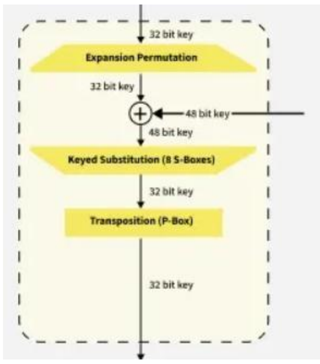
$IP_{broadcast} = 172.16.1.47\ /28$

Subnet/1 = $IP_{net} = 172.16.1.48\ /28$

$IP_{usable} = 172.16.1.49\ /28 \rightarrow 172.16.1.62\ /28$

$IP_{broadcast} = 172.16.1.63\ /28$

2. Since inside the block cipher you have to do an XOR operation between the results of expansion box and the schedule key:



- Results of expansion box: 001111100001111111000000011111111101110001100011
- The schedule key is: 101010100101010111011010001011100000110101010101

We get a 48 bits of:

- 100101 000100 101000 011010 010100 011101 000100 110110

We divide it into row/column, using the first and last digit of every 6bits in the 8 divided sections of our 48 bits: [11,0010],[00,0010],[10,0100],[00,1010],[01,1110],[00,0010],[10,1011]

We get row,column of [3,2],[0,2],[2,4],[0,13],[0,10],[1,14],[0,2],[2,11]

- The provided fix table:

[5, 6, 1, 13, 14, 0, 6, 12, 1, 15, 7, 9, 14, 2, 7, 3]

[2, 3, 1, 1, 1, 3, 1, 11, 7, 9, 3, 5, 15, 14, 8, 13]

[9, 0, 13, 11, 7, 15, 15, 3, 13, 9, 13, 12, 7, 14, 15, 8]

[4, 12, 4, 12, 13, 6, 13, 2, 14, 4, 6, 8, 14, 1, 9, 8]

After substituting that into our fix table we get [4,1,7,2,7,8,1,12] which in binary is:

[0100 0001 0111 0010 0111 1000 0001 1100]

Thus, the results is 01000001011100100111100000011100.